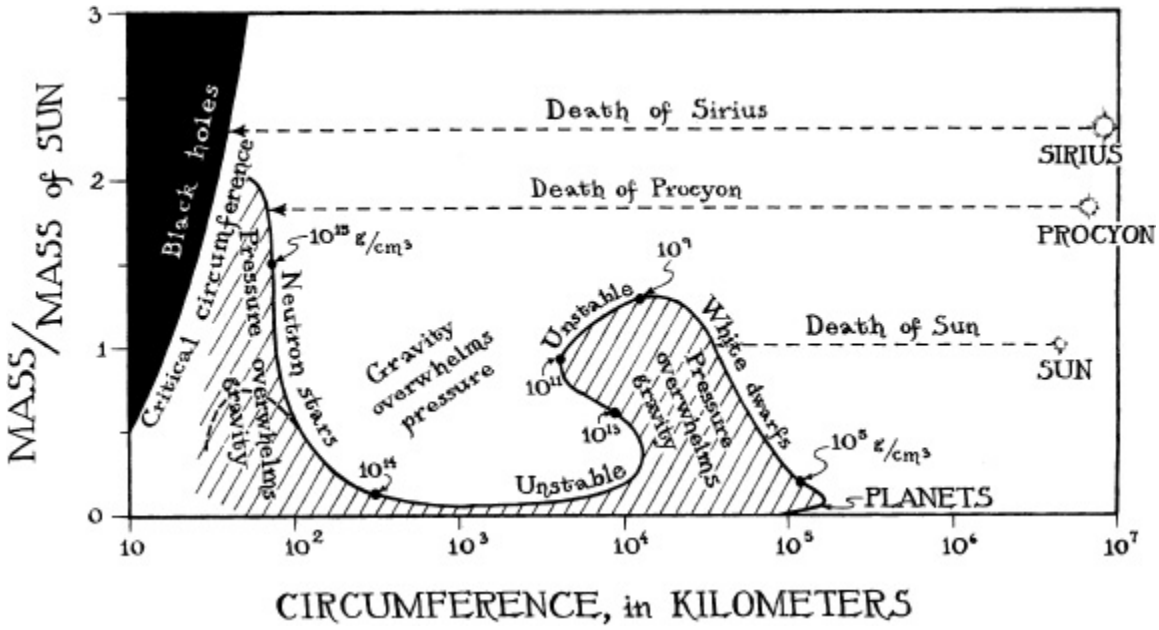
rightward in the diagram. Only along the boundary of white and shaded do gravity and pressure balance each other perfectly; thus, the boundary curve is the curve of cold, dead stars that are in pressure/gravity equilibrium.



5.5 The circumferences (plotted horizontally), masses (plotted vertically), and central densities (labeled on curve) for cold, dead stars, as computed by Masami Wakano under the direction of John Wheeler, using the equation of state of Box 5.5. At central densities above those of an atomic nucleus (above 2 x 1014 grams per cubic centimeter), the solid curve is a modern, 1990s, one that takes proper account of the nuclear force, and the dashed curve is that of Oppenheimer and Volkoff without nuclear forces.

As one moves along this *equilibrium curve*, one is tracing out dead "stars" of higher and higher densities. At the lowest densities (along the bottom edge of the figure and largely hidden from view), these "stars" are not stars at all; rather, they are cold planets made of iron. (When Jupiter ultimately exhausts its internal supply of radioactive heat and cools off, although it is made mostly of hydrogen rather than iron, it will nevertheless lie near the rightmost point on the equilibrium curve.) At higher densities than the planets are Chandrasekhar's white dwarfs.

When one reaches the topmost point on the white-dwarf part of the curve (the white dwarf with Chandrasekhar's maximum mass of 1.4 Suns[7]) and then moves on to still higher densities, one meets cold, dead stars that cannot exist in nature because they are unstable against implosion or explosion (Box 5.6). As one moves from white-dwarf densities toward neutron-star densities, the masses of these unstable equilibrium stars